Chapter 5

Taylor series and Maclaurin Series

The Taylor series of a function $f(x)$ that is infinitely differentiable at $x = a$ is the power series

$$f(x) = f(a) + (x-a)\frac{f'(a)}{1!} + (x-a)^2\frac{f''(a)}{2!} + \cdots$$

which can be written as

$$f(x) = \sum_{k=0}^{\infty} (x-a)^k \frac{f^{(k)}(a)}{k!},$$

where $k!$ denotes the factorial of k and $f^{(k)}(a)$ denotes the k-th derivative of $f(x)$ evaluated at the point $x = a$.

Example 3: Find the first three non zero terms in the expansion of the Taylor series of $f(x) = \sin 2x$ about $x = \frac{\pi}{4}$.

Solution:

$f(x) = \sin 2x$	$f\left(\frac{\pi}{4}\right) = 1$
$f'(x) = 2\cos 2x$	$f'\left(\frac{\pi}{4}\right) = 0$
$f''(x) = -4\sin 2x$	$f''\left(\frac{\pi}{4}\right) = -4$
$f'''(x) = -8\cos 2x$	$f'''\left(\frac{\pi}{4}\right) = 0$
$f^{iv}(x) = 16\sin 2x$	$f^{\text{iv}}\left(\frac{\pi}{4}\right) = 16$

So, $f(x) = \sin 2x = 1 - 4\frac{\left(x-\frac{\pi}{4}\right)^2}{2!} + 16\frac{\left(x-\frac{\pi}{4}\right)^4}{4!} - \cdots = 1 - 2\left(x-\frac{\pi}{4}\right)^2 + 2\left(x-\frac{\pi}{4}\right)^4 - \cdots$

Sample Exercise

1. Expand in Taylor Series/ Maclaurin Series for the following functions:
   (a) $f(x) = \sin x, \sin 2x, \sin 3x, \cos x, \cos 2x, \cos 3x$ at $x = \frac{\pi}{2}$
   (b) $f(x) = \sin x, \sin 2x, \sin 3x, \cos x, \cos 2x, \cos 3x$ at $x = 0$
   (c) $f(x) = e^x, e^{2x}, e^{3x}$ at $x = 0,2,3$ etc.

Applications of Differentiation

- ✓ **Lecture Content:**
- ✓ 4.1 (Page 276) : Maximum Minimum Values
  Definition (Critical Number): Example 7
  Example 8, Example 9

Exercise 4.1

4.1 Exercise-29,30,31,32, 47,48,49,50,51

- ✓ **Lecture Content:**
  4.3: Increasing Decreasing Test: Example 1
  Local Extreme Values: The First derivative Test, Example 3
  Concavity Test, Inflection Point, Second derivative test, Example 6, Example 7

Exercise 4.3

4.3 Exercise-9,10,11, 22

- ✓ **Lecture Content:**
  4.7 Optimization Problems: Example 1,2,3,4,5

Sample Content

Stationary point: If $\frac{dy}{dx} = 0$ at some point of the curve $y = f(x)$, then tangent to the graph is horizontal there. Such a point is called a **stationary point (**or **turning point)**.

Critical Point: A point which is either a stationary point or a point at which it is not differentiable is a **critical point**.

Example:

Find the critical points of the function $f(x) = 5x^2 + 4x - 2$

First, we take the derivative:

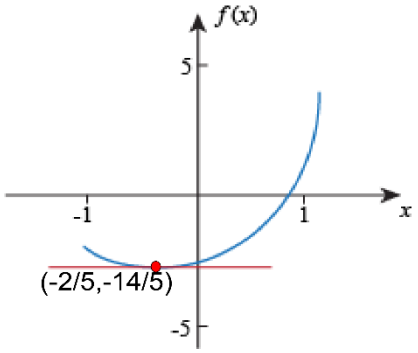
$f'(x) = 10x + 4$.

Since $f'(x)$ is defined everywhere, we can move right on to finding the roots of $f'(x)$. We need to solve the equation

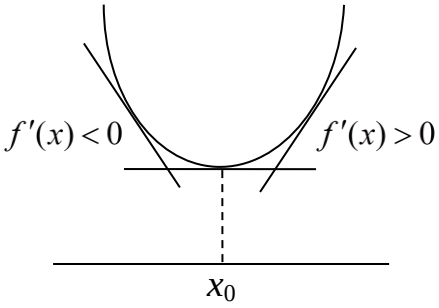
For critical point, $f'(x) = 0$

$$\Rightarrow 10x + 4 = 0$$
$$\Rightarrow 10x = -4$$
$$\Rightarrow x = -\frac{2}{5}$$

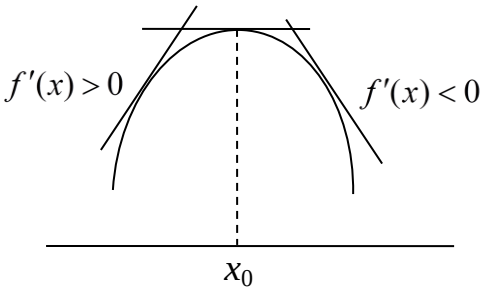
This means -2/5 is a critical point in fact. If we look at the graph, this makes sense, so long as we zoom in enough to see what the graph looks like around x = -2/5:



Maximum and Minimum: A point x_0 is a (relative or local) maximum of $f(x)$ if $f(x_0) \geq f(x)$ for all x sufficiently near x_0 and x_0 is a (relative or local) minimum of $f(x)$ it $f(x_0) \leq f(x)$.



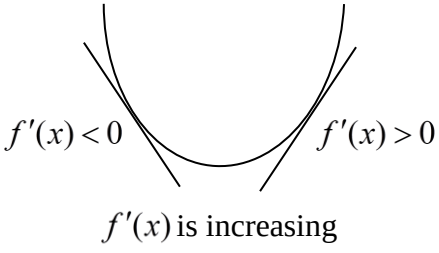
A Local Minimum



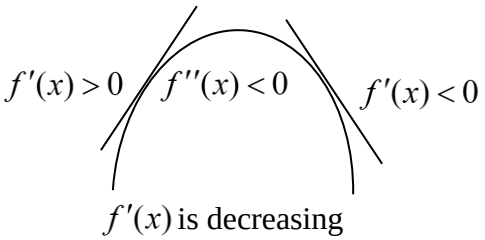
A Local Maximum

To find the nature of the relative extrema (maxima or minima) we can use the signs of the first derivative on either side of the stationary points.

Concavity: Let $f(x)$ be differentiable on some interval $I = (a,b)$. Then $f(x)$ is concave upwards if $f'(x)$ is increasing on I. It is concave downwards if $f'(x)$ is decreasing. Provided that $f(x)$ is twice differentiable, then $f'(x)$ is increasing if $f''(x) > 0$. Hence if $f''(x_0) > 0$ then $f(x)$ is concave upwards near x_0, while if $f''(x_0) < 0$ then $f(x)$ is concave downwards near x_0.

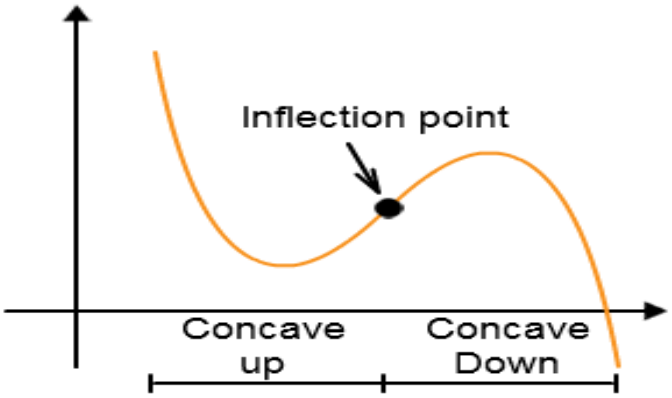


Concave Upwards



Concave Downwards

Inflection Point: Inflection points are the points where the function changes concavity. A point x_0 is a point of inflection of $f(x)$ if $f''(x)$ changes sign via x_0.



Nature of Relative Extrema

First Derivative Test: Suppose $f(x)$ is differentiable and $f'(x_0)=0$. Then

(1) If $f'(x)$ changes sign from positive to negative at x_0 then $f(x)$ has a relative maximum at x_0.

(2) If $f'(x)$ changes sign from negative to positive at x_0 then $f(x)$ has a relative minimum at x_0.

Second Derivative Test: Suppose f is twice differentiable. Then

(1) If $f'(x_0)=0$ and $f''(x_0)<0$ then $f(x)$ has a maximum at x_0.

(2) If $f'(x_0)=0$ and $f''(x_0)>0$ then $f(x)$ has a minimum at x_0.

(3) If $f'(x_0)=0$ and $f''(x_0)=0$ then the second derivative test is inconclusive.

Sample Example 1: Consider the function $f(x)=x^3+3x^2-9x+1$, then

(a) Find the stationary point(s).

(b) Find the interval(s) on which the function is increasing and decreasing.

(c) Find the point(s) of inflection.

(d) Find the interval(s) on which the function is concave up and concave down.

(e) Discuss the nature of stationary point(s).

(f) Sketch the graph.

Solution: Given that $f(x)=x^3+3x^2-9x+1$

(a) $f'(x)=3x^2+6x-9$

At the stationary points, $f'(x)=3x^2+6x-9=0 \Rightarrow x=-3,1$

Now, $f(-3)=28$ and $f(1)=-4$. So, the stationary points are (-3, 28) and (1, -4).

(b) $f'(x)=3x^2+6x-9$

Intervals	Sign of $f'(x)=3x^2+6x-9$	Comment

$(-\infty,-3)$	Positive	*f(x)* is increasing in $(-\infty,-3)$
$(-3,1)$	Negative	*f(x)* is decreasing in $(-3,1)$
$(1,+\infty)$	Positive	*f(x)* is increasing in $(1,+\infty)$

(c) $f''(x)=6x+6$

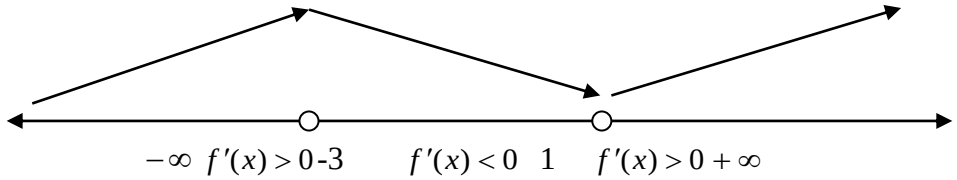
At the points of inflection, $f''(x)=6x+6=0, \Rightarrow x=-1$

Now, $f(-1)=12$ So, the point of inflection is (-1, 12).

(d) $f''(x)=6x+6$

Intervals	Sign of $f''(x)=6x+6$	Comment
$(-\infty,-1)$	negative	*f(x)* is concave down in $(-\infty,-1)$
$(-1,+\infty)$	positive	*f(x)* is concave up in $(-1,+\infty)$

(e)



From the above figure, $f(x)$ has a local maxima at (-3, 28) and local minima at (1, -4).

Sample Applications:

Example 2: The perimeter of a rectangular field is 100 meters. Find the maximum possible area and the dimensions of the field in this case.

Solution: Let, the length of the rectangular field is *x* m and the breadth is *y* m.
So, the area is *xy* square meter.

Perimeter, $2(x+y)=100 \Rightarrow x+y=50 \Rightarrow y=50-x$

So, $f(x)=x(50-x)=50x-x^2$

Now, $f'(x)=50-2x$, $f''(x)=-2$

For extremum area, $f'(x)=50-2x=0, \Rightarrow x=25$

At $x=25$, $f''(x)=-2<0$ which indicates that $f(x)$ has a maxima at $x=25$.

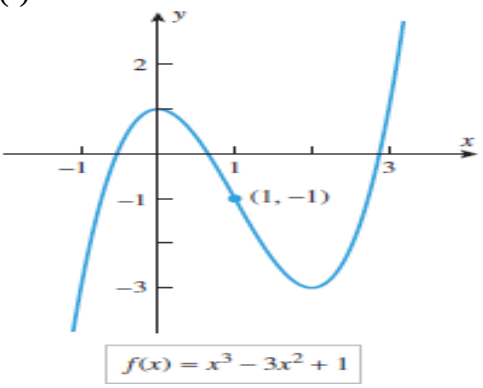
So, $y = 50 - 25 = 25$ and maximum area = (25)(25) = 625 square meter.

Sample Exercises 5.1

1. (i) $f(x)=x^3-3x^2+1$,(ii) $f(x)=x^3-x^2-5x-5$,(iii) $f(x)=x+2\sin x$,where $0<x<2\pi$

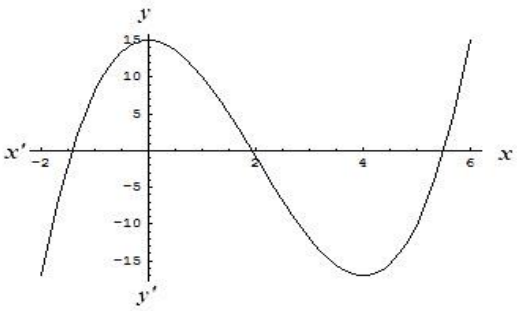
(a) find the stationary point(s),
(b) find the interval (s) on which the function is increasing and decreasing,
(c) find the point(s) of inflection (if any),
(d) find the interval (s) on which the function is concave up and concave down,
(e) discuss the nature of stationary point(s) and
(f) sketch the graph showing all of the above information's.

Answer: 1. (i) (a) $0, 2$, (b) increasing on $(-\infty, 0), (2, \infty)$; decreasing on $(0, 2)$, (c) inflection point $x = 1$, (d) concave up $(1, \infty)$ (e) minimum value at $x = 0$ and no maximum value at $x = 2$, (f)

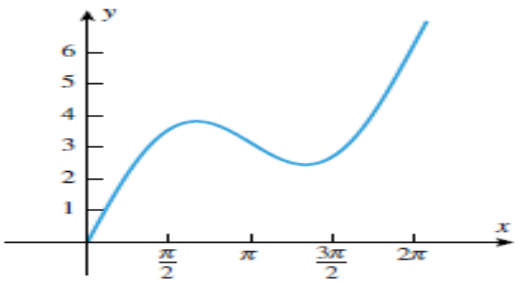


$f(x) = x^3 - 3x^2 + 1$

(ii) (a) $-\frac{5}{3}$ and 4; (b) increasing on $\left(-\infty, -\frac{5}{3}\right)$ and $(1, +\infty)$; decreasing on $\left(-\frac{5}{3}, 1\right)$, (c) $-\frac{1}{3}$, (d) concave up on $\left(-\frac{1}{3}, \infty\right)$; concave down on $\left(-\infty, -\frac{1}{3}\right)$, (e) maximum at $x = -\frac{5}{3}$, minimum at $x = 1$, (f)

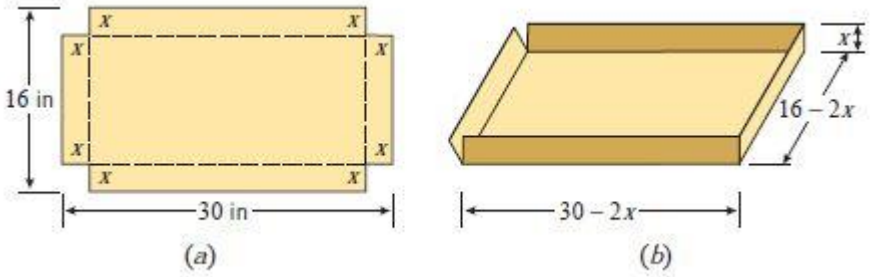


iii) (a) $\frac{2\pi}{3}, \frac{4\pi}{3}$,(b) increasing on $\left(0, \frac{2\pi}{3}\right)$ and $\left(\frac{4\pi}{3}, 2\pi\right)$, decreasing on $\left(\frac{2\pi}{3}, \frac{4\pi}{3}\right)$, (c) concave down (0,π) and concave up (π,2π), (e) maximum at $x = \frac{2\pi}{3}$ and minimum at $x = \frac{4\pi}{3}$.
f)



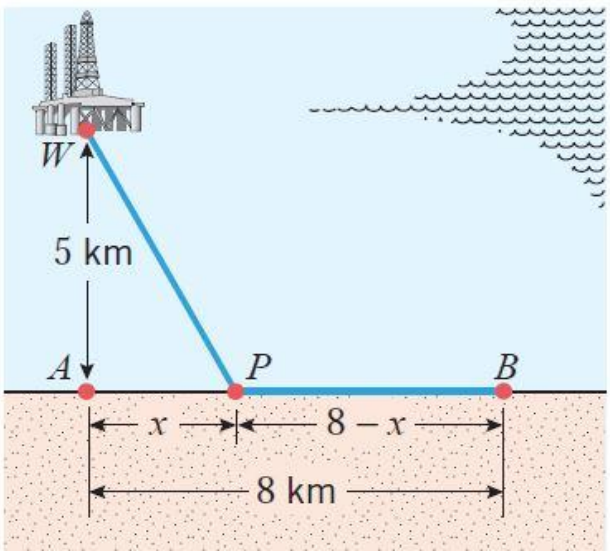
$f(x) = x + 2\sin x$

3. An open box is to be made from a 16 inch by 30 inch piece of cardboard by cutting out squares of equal size from the four corners and bending up the sides. What size should the squares be to obtain a box with the largest volume?



Answer: Maximum square size should be $\frac{10}{3}$ inches with the greatest possible volume $\approx 726in^3$

3.The following figure shows an offshore oil well located at a point W that is $5\ km$ from the closest point A on the straight shoreline. Oil is to be piped from W to a shore point B that is $8\ km$ from A by piping it on a straight line under water from W to some shore point P



between A and B and then on to B via pipe along the shoreline. If the cost of laying pipe is 1million/km under water and $.5\ million/km$ over land, where should the point P be located to minimize the cost of laying the pipe?

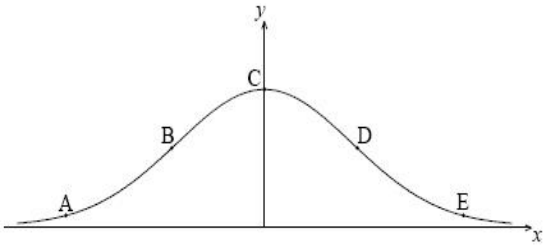
Answer: The point P should be located $\approx 2.89km$ from A with least possible cost 8.330127 million $.

4.A particle moves along a straight line so that its velocity, $p\ ms^{-1}$, at time *s* seconds is given by $p = 120 + 10s - 5s^2$, for $0 \leq s \leq 6$

(i) Find the value of *s* when the speed of the particle is greatest.

(ii) Find the acceleration of the particle when its speed is zero

5.The following diagram shows the graph of $f(x) = e^{-x^2}$.



The points A, B, C, D and E lie on the graph of f . Two of these are points of inflexion.

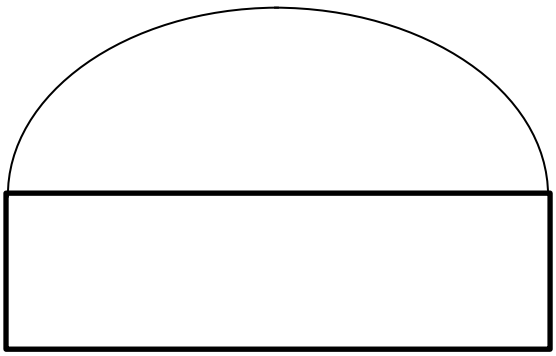
(a) Identify the **two** points of inflexion.

(b) (i) Find $f'(x)$, (ii) Show that $f''(x) = (4x^2 - 2)e^{-x^2}$.

(c) Find the *x*-coordinate of each point of inflexion.

(d) Use the second derivative to show that one of these points is a point of inflexion.

6.



Find the dimension of the rectangle that will maximize the area of the whole region if the perimeter is 20 ft.